

Estimated Annual Agricultural Pesticide Use Hood River County, Oregon 2014-2018

24 July 2026

Summary

Hood River County ranks **#4 of Oregon's 36 counties** by total estimated agricultural pesticide mass applied (EPest-high method, summed 2014-2018), making it one of OR's five highest pesticide-use counties. Hood River County, in the Hood River Valley, is nationally known for its pear and apple orchards.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Hood River County, Oregon** during 2018, as reported by [USGS Estimated Annual Agricultural Pesticide Use](#) data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound's chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by [Thelin and Stone \(2013\)](#) and [Baker and Stone \(2015\)](#):

“EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD.” -[Pesticide National Synthesis Project webpage](#)

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. “detailed interpretation of where and how much use occurs within a county is not appropriate”).
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound
OTH	Other

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)
(continued)

Code	Chemical class
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Hood River County, OR, 1-Yr (2018),
3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification
key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	PETROLEUM OIL	OTH	360520	515704	512041
2	SULFUR	INO	135585	146914	161126
3	MANCOZEB	DTC	56411	57487	58237
4	COPPER	INO	11779	8191	8776
5	GLYPHOSATE	OTH	8270	6203	6424
6	CALCIUM POLYSULFIDE	INO	5512	25264	24723
7	CHLORPYRIFOS	OP	4133	2942	2441
8	MALATHION	OP	3764	1700	1092
9	2,4-D	CPX	3362	4261	4885
10	DIURON	OTH	3044	3437	3110
11	THIOPHANATE-METHYL	OTH	2996	2460	2470
12	COPPER SULFATE	INO	2980	11329	10556
13	PYRIDABEN	OTH	2645	3540	3209
14	TRIFLUMIZOLE	OTH	1944	2062	2112
15	KAOLIN CLAY	INO	1643	229107	191370
16	PENDIMETHALIN	OTH	1477	1939	2850
17	NOVALURON	OTH	1301	1106	869
18	BACILLUS THURINGIENSIS	MIC	1070	766	592
19	MCPA	CPX	1059	538	649
20	COPPER HYDROXIDE	INO	1025	1467	1547
21	ACETAMIPRID	OTH	920	458	358
22	COPPER OXYCHLORIDE	INO	838	289	175
23	SETHOXYDIM	OTH	829	369	223
24	GLUFOSINATE	OTH	815	497	365
25	SPIROTETRAMAT	OTH	770	1084	1206
26	BUPROFEZIN	OTH	744	817	1141
27	PARAQUAT	DIP	608	609	473
28	IMIDACLOPRID	OTH	600	418	382
29	PENTHIOPYRAD	OTH	533	958	1198
30	NORFLURAZON	OTH	512	603	896
31	PYRIPROXYFEN	OTH	501	604	574
32	BACILLUS SUBTILIS	MIC	492	803	544
33	ETOXAZOLE	OTH	465	503	423
34	OXYTETRACYCLINE	OTH	437	971	740
35	FLUOPYRAM	OTH	434	270	174
36	TRIFLOXYSTROBIN	OTH	397	263	195
37	SPIRODICLOFEN	OTH	393	258	251
38	POTASSIUM BICARBONATE	INO	326	1061	723
39	CHLORANTRANILIPROLE	OTH	317	427	394
40	CARBARYL	CB	314	322	440
41	THIAMETHOXAM	OTH	296	162	155
42	CYFLUMETOFEN	OTH	286	213	168
43	ORYZALIN	OTH	267	2892	2386
44	CYHALOTHRIN-LAMBDA	PYR	247	210	176
45	STREPTOMYCIN	OTH	247	701	500
46	INDAZIFLAM	OTH	220	149	163
47	PYRACLOSTROBIN	OTH	206	402	471

Table 2: Top 80 EPest-high Estimates, Hood River County, OR, 1-Yr (2018),
3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification
key table above) (*continued*)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
48	FLUTRIAFOL	OTH	188	74	54
49	CYANTRANILIPROLE	OTH	181	76	60
50	METAM	DTC	174	172	242
51	BOSCALID	OTH	171	551	699
52	SPINOSYN	OTH	170	210	163
53	SPINETORAM	OTH	155	519	462
54	RIMSULFURON	OTH	150	183	150
55	QUINOXYFEN	OTH	141	91	108
56	DIAZINON	OP	137	406	283
57	NAPHTHYLACETIC ACID	OTH	132	139	149
58	FLUXAPYROXAD	OTH	116	106	64
59	BURKHOLDERIA SPP	MIC	99	75	72
60	CUPROUS OXIDE	INO	97	211	390
61	TRIFLURALIN	OTH	89	72	108
62	FENPYROXIMATE	OTH	87	120	106
63	BACILLUS AMYLOLIQUEFACIEN	MIC	86	352	352
64	ABAMECTIN	OTH	83	140	139
65	CHLOROPICRIN	OC	75	32	150
66	FENPROPATHRIN	PYR	74	141	258
67	MYCLOBUTANIL	OTH	70	75	84
68	PROHEXADIONE	OTH	70	46	41
69	DICHLOROPROPENE	OC	62	107	620
70	AVIGLYCINE	OTH	56	42	51
71	HYDROGEN PEROXIDE	INO	52	31	72
72	SIMAZINE	TRZ	51	167	934
73	AUREOBASIDIUM PULLULANS	MIC	50	599	485
74	CYTOKININ	OTH	48	45	34
75	CAPTAN	OTH	48	4532	6274
76	GIBBERELIC ACID	OTH	46	28	29
77	TEBUCONAZOLE	OTH	40	23	16
78	ATRAZINE	TRZ	34	40	30
79	KASUGAMYCIN	OTH	34	106	106
80	ETHEPHON	OTH	32	31	50

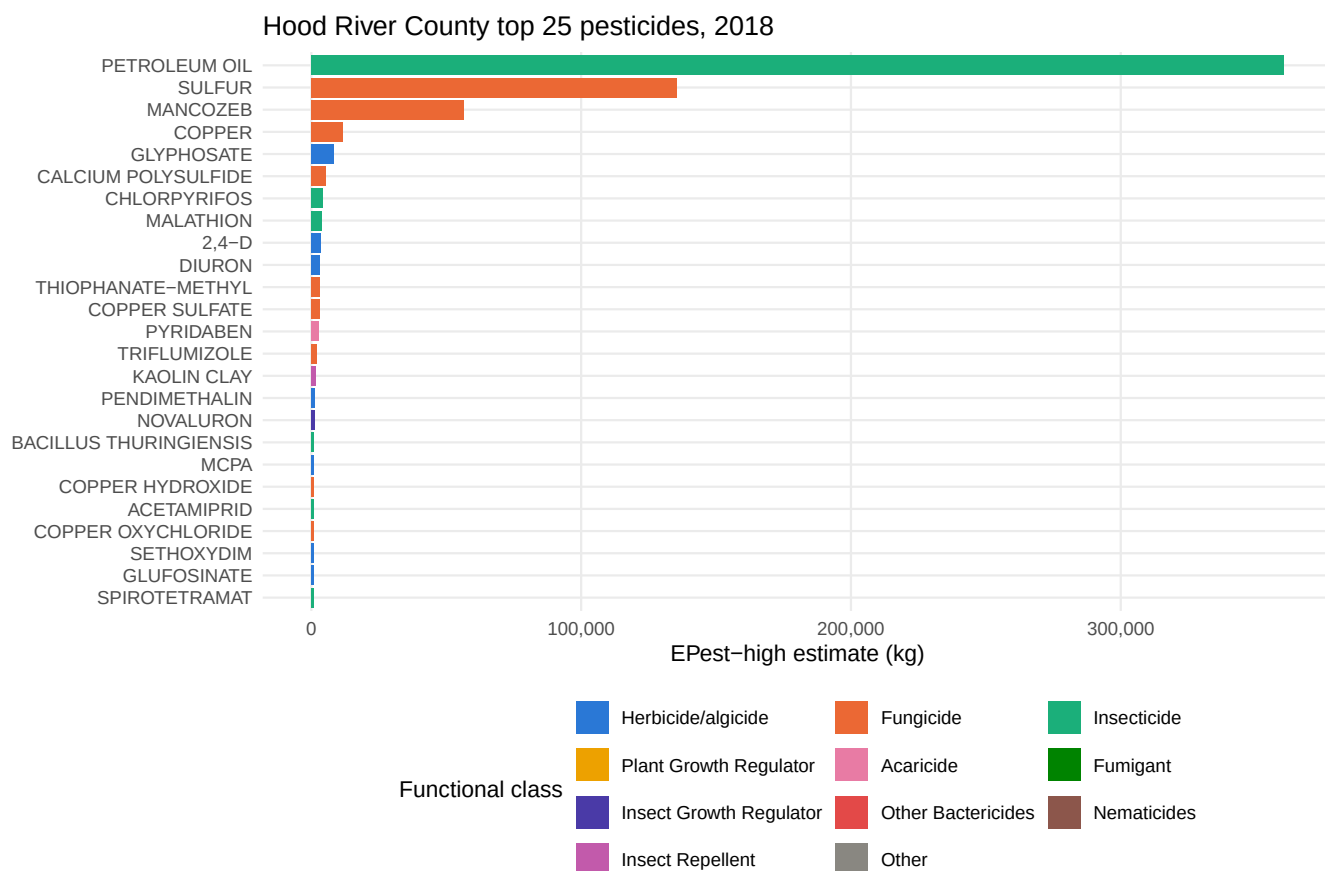


Figure 1: Top 25 compounds in Hood River County by E Pest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Hood River County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	PETROLEUM OIL	360520	360520
2	SULFUR	135585	135585
3	MANCOZEB	56411	56411
4	COPPER	11779	11779
5	GLYPHOSATE	8269	8270
6	CALCIUM POLYSULFIDE	5512	5512
7	CHLORPYRIFOS	1317	4133
8	MALATHION	170	3764
9	2,4-D	3359	3362
10	DIURON	3043	3044
11	THIOPHANATE-METHYL	2996	2996
12	COPPER SULFATE	2980	2980
13	PYRIDABEN	2645	2645
14	TRIFLUMIZOLE	1944	1944
15	KAOLIN CLAY	1643	1643
16	PENDIMETHALIN	1462	1477
17	NOVALURON	1	1301
18	BACILLUS THURINGIENSIS	1070	1070
19	MCPA	1057	1059
20	COPPER HYDROXIDE	918	1025
21	ACETAMIPRID	74	920
22	COPPER OXYCHLORIDE	838	838
23	SETHOXYDIM	829	829
24	GLUFOSINATE	815	815
25	SPIROTETRAMAT	770	770
26	BUPROFEZIN	541	744
27	PARAQUAT	572	608
28	IMIDACLOPRID	117	600
29	PENTHIOPYRAD	533	533
30	NORFLURAZON	512	512
31	PYRIPROXYFEN	501	501
32	BACILLUS SUBTILIS	492	492
33	ETOXAZOLE	465	465
34	OXYTETRACYCLINE	437	437
35	FLUOPYRAM	434	434
36	TRIFLOXYSTROBIN	397	397
37	SPIRODICLOFEN	393	393
38	POTASSIUM BICARBONATE	326	326
39	CHLORANTRANILIPROLE	317	317
40	CARBARYL	314	314
41	THIAMETHOXAM	0	296
42	CYFLUMETOFEN	4	286
43	ORYZALIN	267	267
44	CYHALOTHRIN-LAMBDA	27	247
45	STREPTOMYCIN	247	247
46	INDAZIFLAM	220	220
47	PYRACLOSTROBIN	201	206
48	FLUTRIAFOL	42	188
49	CYANTRANILIPROLE	0	181
50	METAM	111	174

Table 4: Top 50 Hood River County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	PETROLEUM OIL	515704	515704
2	KAOLIN CLAY	61789	229107
3	SULFUR	146914	146914
4	MANCOZEB	57487	57487
5	CALCIUM POLYSULFIDE	25264	25264
6	COPPER SULFATE	11329	11329
7	COPPER	8191	8191
8	GLYPHOSATE	6189	6203
9	CAPTAN	54	4532
10	2,4-D	4252	4261
11	PYRIDABEN	3540	3540
12	DIURON	3428	3437
13	CHLORPYRIFOS	1315	2942
14	ORYZALIN	2758	2892
15	THIOPHANATE-METHYL	2087	2460
16	TRIFLUMIZOLE	2062	2062
17	PENDIMETHALIN	1922	1939
18	MALATHION	177	1700
19	COPPER HYDROXIDE	903	1467
20	NOVALURON	287	1106
21	SPIROTETRAMAT	1084	1084
22	POTASSIUM BICARBONATE	1061	1061
23	ZIRAM	132	1059
24	OXYTETRACYCLINE	971	971
25	PENTHIOPYRAD	958	958
26	BUPROFEZIN	750	817
27	ZINC	816	816
28	BACILLUS SUBTILIS	803	803
29	BACILLUS THURINGIENSIS	487	766
30	STREPTOMYCIN	701	701
31	OXYFLUORFEN	677	677
32	PARQUAT	592	609
33	PYRIPROXYFEN	604	604
34	NORFLURAZON	435	603
35	AUREOBASIDIUM PULLULANS	29	599
36	BOSCALID	551	551
37	MCPA	459	538
38	SPINETORAM	519	519
39	ETOXAZOLE	503	503
40	GLUFOSINATE	497	497
41	ACETAMIPRID	45	458
42	CHLORANTRANILIPROLE	427	427
43	IMIDACLOPRID	88	418
44	DIAZINON	145	406
45	PYRACLOSTROBIN	400	402
46	SETHOXYDIM	369	369
47	BACILLUS AMYLOLIQUEFACIEN	352	352
48	CARBARYL	322	322
49	PHOSMET	104	313
50	COPPER OXYCHLORIDE	280	289

Table 5: Top 50 Hood River County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	PETROLEUM OIL	512041	512041
2	KAOLIN CLAY	90959	191370
3	SULFUR	161126	161126
4	MANCOZEB	58231	58237
5	CALCIUM POLYSULFIDE	24269	24723
6	COPPER SULFATE	10517	10556
7	COPPER	8556	8776
8	GLYPHOSATE	6408	6424
9	CAPTAN	49	6274
10	2,4-D	4879	4885
11	PYRIDABEN	3209	3209
12	DIURON	3093	3110
13	PENDIMETHALIN	2430	2850
14	ZIRAM	66	2656
15	THIOPHANATE-METHYL	2246	2470
16	CHLORPYRIFOS	1395	2441
17	ORYZALIN	2306	2386
18	TRIFLUMIZOLE	2112	2112
19	COPPER HYDROXIDE	545	1547
20	SPIROTETRAMAT	1205	1206
21	PENTHIOPYRAD	1198	1198
22	BUPROFEZIN	682	1141
23	MALATHION	171	1092
24	OXYFLUORFEN	937	945
25	SIMAZINE	895	934
26	NORFLURAZON	261	896
27	NOVALURON	176	869
28	PHOSMET	76	852
29	OXYTETRACYCLINE	731	740
30	POTASSIUM BICARBONATE	674	723
31	BOSCALID	699	699
32	MCPA	278	649
33	DICHLOROPROPENE	160	620
34	BACILLUS THURINGIENSIS	309	592
35	PYRIPROXYFEN	572	574
36	ZINC	288	565
37	BACILLUS SUBTILIS	537	544
38	STREPTOMYCIN	490	500
39	AUREOBASIDIUM PULLULANS	17	485
40	PARAQUAT	448	473
41	PYRACLOSTROBIN	469	471
42	SPINETORAM	462	462
43	CARBARYL	422	440
44	ETOXAZOLE	423	423
45	CHLORANTRANILIPROLE	394	394
46	CUPROUS OXIDE	256	390
47	IMIDACLOPRID	110	382
48	GLUFOSINATE	298	365
49	CYPRODINIL	357	364
50	ACETAMIPRID	34	358